PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

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AI "Failure"

What does it mean for AI to "fail"?

- Predictive: Gets things "wrong"
- Generative: Gives the user something other than what they want
- Misrepresents "reality"
- Causes / exacerbates harm

Bias, Unfairness, and Harm

Bias: treating members of some groups differently based on irrelevant / ungermane criteria.

Unfairness: making predictions / generating content that harms (some) people without transparency or human review.

Harm: oppression; denial of property / liberty / opportunity; misrepresentation; exacerbating existing inequality; etc.

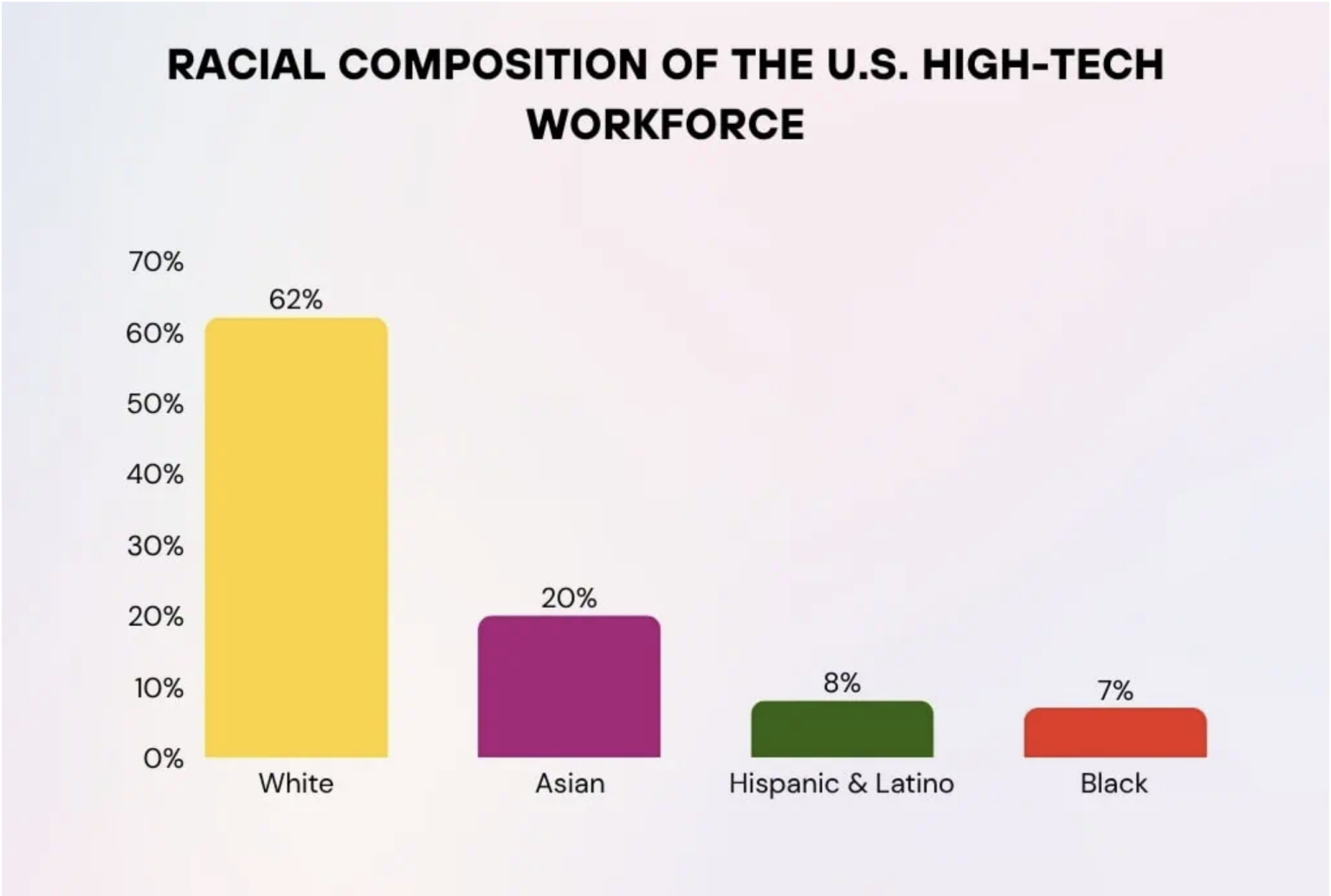
The Tech Industry

Gender & Sexuality

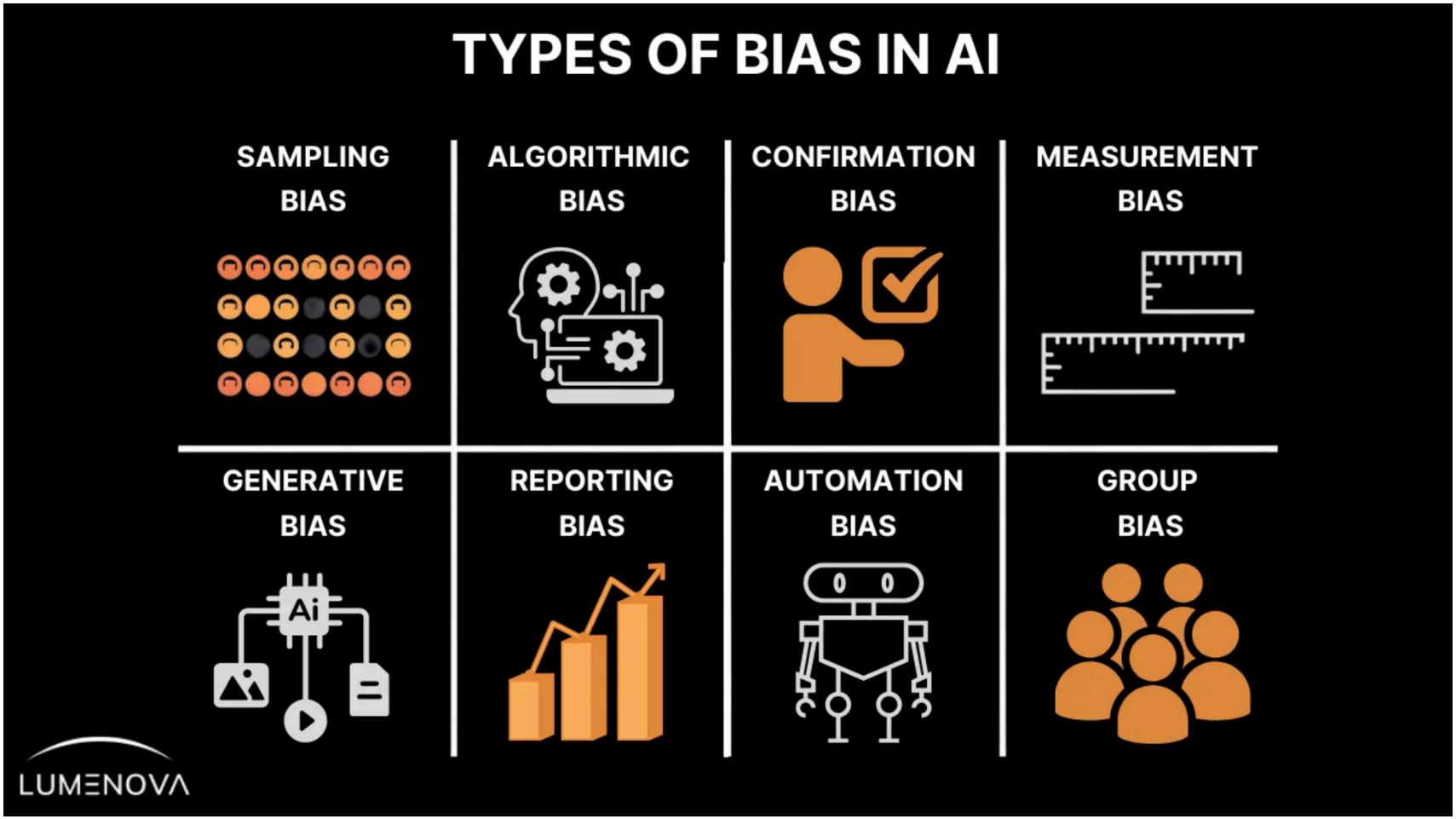
- Women are $\approx 27\%$ of the tech workforce, $< 20\%$ of leadership
  - Highest in UX/UI design (46%)
  - Lowest in cybersecurity (12%)
- Female burnout rate is 57% (vs. 36% for men)
- 21% of CS degrees awarded to women (vs. 37% in 1984)
- Half of all women who enter tech leave before age 35
- 2-3% of tech workers openly identify as LBGTQ+
- 65% of LGBTQ+ tech employees report verbal harassment / aggressions

(source one, source two)

Race and Ethnicity



(source)


TYPES OF BIAS IN AI
SAMPLING BIAS
ALGORITHMIC BIAS
CONFIRMATION BIAS
MEASUREMENT BIAS
GENERATIVE BIAS
Ai
REPORTING BIAS
AUTOMATION BIAS
GROUP BIAS
LUMENOVA

(source)

Narayanan & Kapoor (2024)



(source)

Mechanisms:

- The Data
- Proxies, not outcomes
- Masking / removing traits fails
- Subgroup accuracy
- Feedback loops
- Asymmetrical costs